Title: NLP-Based Analysis of Dialogues in Ukrainian Films

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1. Introduction

Cinema is not only a form of art but also a rich source of structured and unstructured data for statistical and linguistic research. In the digital era, there is a growing need for automated tools to analyze dialogue structures, character representation, and linguistic patterns. This is particularly relevant for Ukrainian cinema, where specialized NLP and multimodal solutions tailored to the domestic media landscape are currently lacking. This research presents a comprehensive automated system designed to process, and analyze movie dialogues, focusing on gender representation, speech dynamics, and lexical markers. The system is validated on a custom-built corpus of 135 Ukrainian films.

2. Methodology and Processing Pipeline

To process raw video files into structured analytical data, I engineered a robust end-to-end processing pipeline consisting of the following sequential stages:

- **Speaker Diarization:** Utilizing **Pyannote.audio** to segment the audio stream and answer the question "who spoke when," effectively separating individual speaker turns.
- **Speech-to-Text (ASR):** Applying the **Whisper** model to transcribe the segmented audio into Ukrainian text.
- **Text Cleaning:** A custom preprocessing module to normalize the transcribed text and remove artifacts.
- **Morphological Analysis:** Using **UDPipe** for tokenization, lemmatization, and part-of-speech (POS) tagging.
- **Gender Extraction & Analysis:** The core analytical engine that determines speaker gender and extracts statistical metrics.

3. Multimodal Gender Identification

A distinguishing feature of this system is its tri-level approach to identifying the gender of the speaker for each dialogue replica, ensuring high accuracy by cross-referencing multiple modalities:

1. **Linguistic Analysis (Text):** Utilizing UDPipe to detect grammatical gender markers (e.g., feminine/masculine verb endings in past tense, adjectives) within the transcribed replicas.
2. **Acoustic Analysis (Audio):** Processing short audio segments of the replicas using an acoustic classification model to predict speaker gender based on voice frequencies and pitch.
3. **Visual Search (Vision):** Using **Grounding DINO** (an Open-Set Object Detector) to analyze the visual frames corresponding to the timestamp of the replica, detecting and classifying the gender of the character currently on screen.

4. Feature Extraction and Analytics

The processed data allows the system to generate a comprehensive statistical profile for any given Ukrainian movie:

- **Time Distribution and Dynamics:** Calculating total speaking time per gender versus silence intervals. It also analyzes the pace of speech across different movie segments and detects extended monologues (replicas lasting ≥ 30 seconds).
- **Gender Representation:** Automating the **Bechdel Test** to evaluate female representation in the plot, alongside basic quantitative metrics (e.g., which gender dominates the dialogue volume).
- **Lexical Profiling:** Forming lists of the most frequently used words broken down by POS (nouns, adjectives, verbs) for each gender.
- **Lexical Markers (TF-IDF):** By treating male and female replicas as two distinct "documents," the system applies the **TF-IDF** metric to mathematically extract unique, gender-specific vocabulary and speech patterns.

5. Conclusion

This multimodal framework serves as a powerful analytical tool for film critics, screenwriters, and producers. It enables objective evaluation of character balance, uncovers hidden linguistic trends, and identifies specific gender markers at both the scriptwriting stage and post-production. By automating this analysis, the project contributes to a deeper, data-driven understanding of the evolution and current state of Ukrainian cinema.

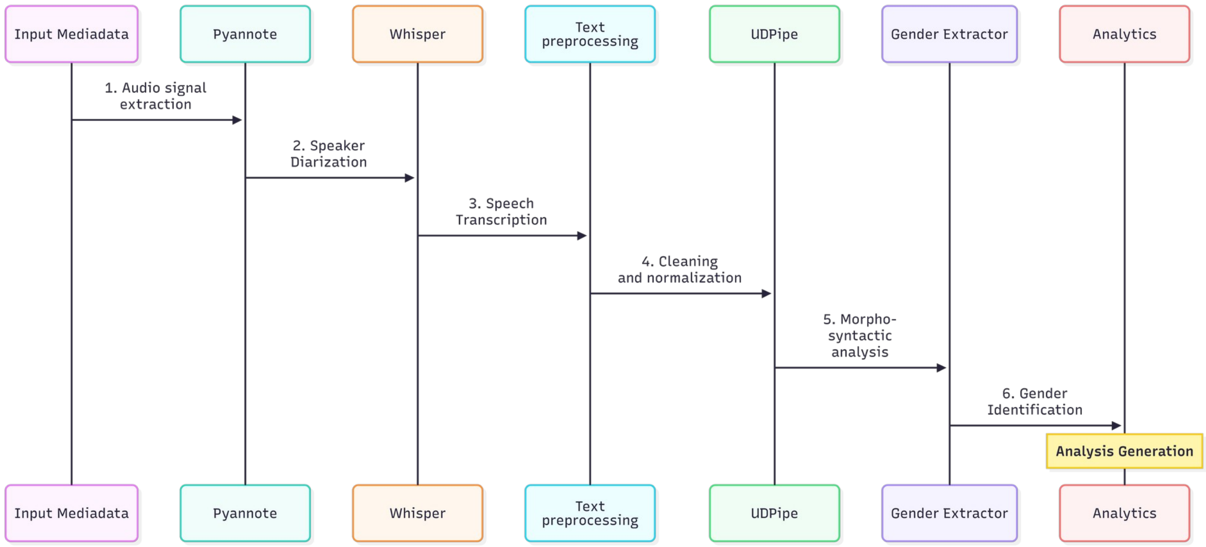

Figure 1: *System architecture and sequential processing pipeline: from raw media input via Pyannote and Whisper to NLP processing (UDPipe) and final multimodal gender extraction*

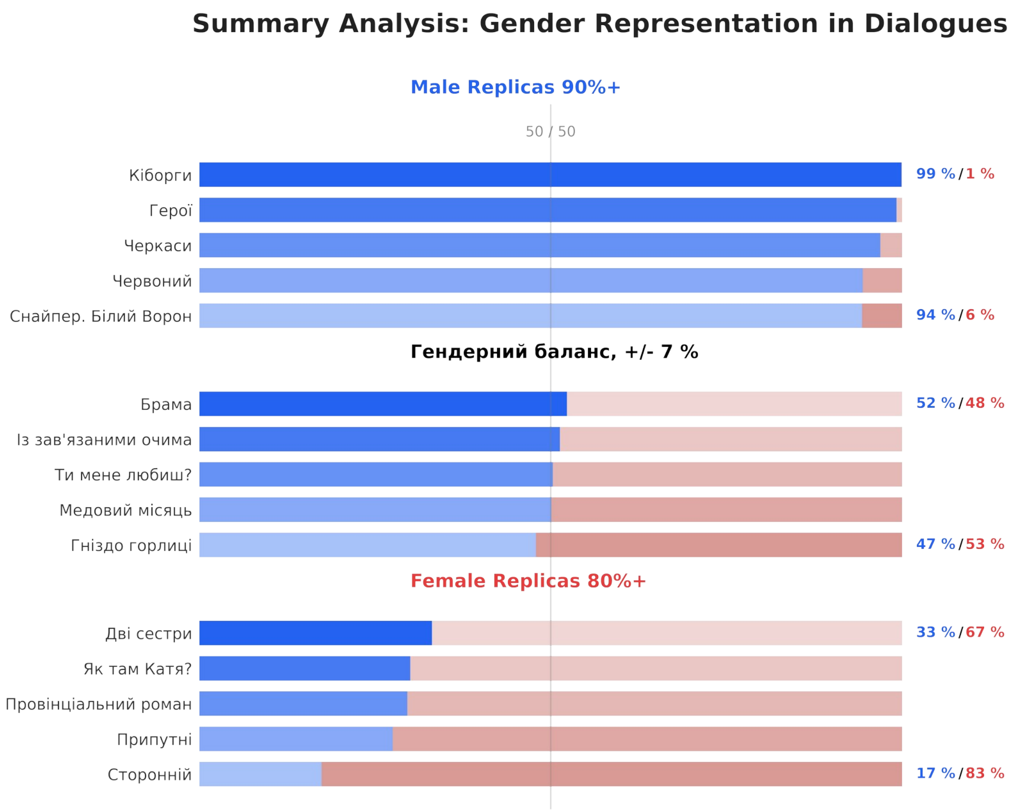

Figure 2: *Summary of dialogue distribution by gender across the analyzed Ukrainian film corpus. Bars illustrate the percentage of replicas for male characters (blue) versus female characters (pink) for each film. The chart categorizes films into three groups: extreme male dominance (≥90%), balanced representations, and significant female presence (≥80%).*

6. References

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